Chapter 5.1 - Memory Hierarchy & Locality Principle

Comprehensive Revision Flashcards • 58 Key Concepts & Practice Questions

#1	**Q1: What illusion do memory hierarchies create for programmers?**
	Answer: They create the illusion of having unlimited amounts of fast memory.

#2	**Q2: What fundamental principle underlies the operation of memory hierarchies?**
	Answer: The principle of locality.

#3	**Q3: What does the principle of locality state about program behavior?**
	Answer: It states that programs access a relatively small portion of their address space at any instant of time.

#4	**Q4: What are the two different types of locality?**
	Answer: Temporal locality (locality in time) and spatial locality (locality in space).

#5	**Q5: Term: Temporal locality**
	Answer: The principle stating that if a data location is referenced, it will tend to be referenced again soon.

#6	**Q6: What common program structure demonstrates high temporal locality?**
	Answer: Loops, where instructions and data are accessed repeatedly.

#7	**Q7: Term: Spatial locality**
	Answer: The principle stating that if a data location is referenced, data locations with nearby addresses will tend to be referenced soon.

#8	**Q8: How does sequential access of instructions in a program demonstrate spatial locality?**
	Answer: Instructions are normally accessed sequentially, meaning that referencing one instruction makes it highly likely the next one in memory will be referenced soon.

#9	**Q9: Sequential access to elements of an array is an example of high _____ locality.**
	Answer: spatial

#10	**Q10: How do computer systems take advantage of the principle of locality?**
	Answer: By implementing the memory of a computer as a memory hierarchy.

#11	**Q11: Term: Memory hierarchy**
	Answer: A structure that uses multiple levels of memories where size and access time increase with distance from the processor.

#12	**Q12: What is the relationship between speed, size, and cost per bit in a memory hierarchy?**
	Answer: Faster memories are more expensive per bit and are therefore smaller.

#13	**Q13: What is the overall goal of implementing a memory hierarchy?**
	Answer: To provide memory as large as the cheapest level, with an access speed offered by the fastest level.

#14	**Q14: In a memory hierarchy, a level closer to the processor is generally a _____ of any level further away.**
	Answer: subset

#15	**Q15: In a memory hierarchy, data is copied between only _____ adjacent levels at a time.**
	Answer: two

#16	**Q16: Term: Block (or line)**
	Answer: The minimum unit of information that can be either present or not present in a level of the memory hierarchy.

#17	**Q17: What is it called when data requested by the processor is found in the upper level of the memory hierarchy?**
	Answer: A hit.

#18	**Q18: What is it called when data requested by the processor is not found in the upper level of the memory hierarchy?**
	Answer: A miss.

#19	**Q19: What action is taken in a memory hierarchy when a miss occurs?**
	Answer: The lower level in the hierarchy is accessed to retrieve the block containing the requested data.

#20	**Q20: Term: Hit rate**
	Answer: The fraction of memory accesses found in a level of the memory hierarchy.

#21	**Q21: Term: Miss rate**
	Answer: The fraction of memory accesses not found in a level of the memory hierarchy.

#22	**Q22: How is the miss rate related to the hit rate?**
	Answer: The miss rate is equal to 1 minus the hit rate.

#23	**Q23: Term: Hit time**
	Answer: The time required to access a level of the memory hierarchy, including the time to determine if the access is a hit or a miss.

#24	**Q24: Term: Miss penalty**
	Answer: The time required to replace a block in an upper level with the corresponding block from the lower level and deliver it.

#25	**Q25: Why is the hit time much smaller than the miss penalty?**
	Answer: Because the upper level is smaller and built with faster memory parts than the lower level.

#26	**Q26: Memory hierarchies keep more recently accessed data closer to the processor to take advantage of _____ locality.**
	Answer: temporal

#27	**Q27: How do memory hierarchies leverage spatial locality?**
	Answer: By moving blocks consisting of multiple contiguous words in memory to upper levels of the hierarchy.

#28	**Q28: In a true memory hierarchy, if data is present in level 'i', where else must it also be present?**
	Answer: It must also be present in the next lower level, 'i + 1'.

#29	**Q29: According to the 'Check Yourself' section, which of the following is generally true: 'Most of the cost of the memory hierarchy is at the highest level.'?**
	Answer: This statement is generally true.

#30	**Q30: According to the 'Check Yourself' section, which of the following is generally true: 'Most of the capacity of the memory hierarchy is at the lowest level.'?**
	Answer: This statement is generally true.

#31	**Q31: In the library analogy for memory hierarchies, what do the books on your desk represent?**
	Answer: The upper, faster level of the memory hierarchy (e.g., a cache).

#32	**Q32: In the library analogy, going back to the shelves to find a book corresponds to what event in a memory hierarchy?**
	Answer: A miss, which requires accessing the lower level of memory.

#33	**Q33: In the library analogy, what does finding the information you need in a book already on your desk represent?**
	Answer: A hit in the memory hierarchy.

#34	**Q34: What fundamental principle states that programs access a relatively small portion of their address space at any instant of time?**
	Answer: The principle of locality.

#35	**Q35: The locality principle stating that if an item is referenced, it will tend to be referenced again soon is known as _____ locality.**
	Answer: temporal

#36	**Q36: What common programming structure results in large amounts of temporal locality?**
	Answer: Loops, which cause instructions and data to be accessed repeatedly.

#37	**Q37: The locality principle stating that if an item is referenced, items whose addresses are close by will tend to be referenced soon is known as _____ locality.**
	Answer: spatial

#38	**Q38: What is an example of program behavior that exhibits high spatial locality?**
	Answer: Sequential access of instructions or sequential access to elements of an array.

#39	**Q39: How do memory hierarchies take advantage of temporal locality?**
	Answer: By keeping more recently accessed data items closer to the processor.

#40	**Q40: How do memory hierarchies take advantage of spatial locality?**
	Answer: By moving blocks of multiple contiguous words to upper levels of the hierarchy.

#41	**Q41: In a memory hierarchy, what happens to the size of the memories and the access time as the distance from the processor increases?**
	Answer: Both the size of the memories and the access time increase.

#42	**Q42: What is the primary goal of using a memory hierarchy?**
	Answer: To provide memory as large as the cheapest technology, with access speeds offered by the fastest memory.

#43	**Q43: What is the relationship between data in a higher level of the memory hierarchy and data in a lower level?**
	Answer: A level closer to the processor is generally a subset of any level further away.

#44	**Q44: In most systems, if data is present in memory level 'i', where else must it be present?**
	Answer: It must also be present in level 'i + 1'.

#45	**Q45: In a memory hierarchy, data is copied between only two _____ levels at a time.**
	Answer: adjacent

#46	**Q46: What is a `hit` in the context of a memory hierarchy?**
	Answer: A memory access where the requested data is found in the upper level of the hierarchy.

#47	**Q47: What is a `miss` in the context of a memory hierarchy?**
	Answer: A memory access where the requested data is not found in the upper level of the hierarchy.

#48	**Q48: What action is taken in a memory hierarchy when a `miss` occurs?**
	Answer: The lower level in the hierarchy is accessed to retrieve the block containing the requested data.

#49	**Q49: How is the `miss rate` related to the `hit rate`?**
	Answer: The miss rate is equal to 1 minus the hit rate (1 - hit rate).

#50	**Q50: What is the major component of the miss penalty?**
	Answer: The time to access the next, lower level in the hierarchy.

#51	**Q51: Why must modern programmers understand the memory hierarchy?**
	Answer: To write code that achieves good performance, as memory access is a major factor.

#52	**Q52: With a high enough hit rate, a memory hierarchy has an effective access time close to that of the _____ level.**
	Answer: highest (and fastest)

#53	**Q53: A memory hierarchy gives the user the illusion of a memory size equal to that of the _____ level.**
	Answer: lowest (and largest)

#54	**Q54: In the library analogy, what do the books on the desk represent?**
	Answer: The upper, faster level of the memory hierarchy (e.g., the cache).

#55	**Q55: In the library analogy, what does going back to the shelves to find a book represent?**
	Answer: A 'miss' that requires accessing the lower, slower level of the memory hierarchy.

#56	**Q56: In the library analogy, what does a single book represent?**
	Answer: A 'block' or 'line' of information.

#57	**Q57: According to the 'Check Yourself' section, which statement is generally true: 'Most of the cost of the memory hierarchy is at the highest level'?**
	Answer: False, the highest level is smallest, so most cost is in the larger, lower levels.

#58	**Q58: According to the 'Check Yourself' section, which statement is generally true: 'Most of the capacity of the memory hierarchy is at the lowest level'?**
	Answer: True, because the lowest levels are the largest.